

TABLE 13

Component Weight and Weight Percent Based on Total Solution Weight				
	Ibu-1 g (wt %)	Ibu-2 g (wt %)	Ibu-3 g (wt %)	Ibu-4 g (wt %)
Ibuprofen	0.6 (36.8)	0.6 (32.3)	0.6 (31.6)	0.6 (31.1)
NaOH	0	0.115 (6.2)	0.135 (7.1)	0.15 (7.8)
Ethanol	0.4 (24.5)	0.4 (21.5)	0.4 (21.1)	0.4 (20.7)
DI water	0.6 (36.8)	0.715 (38.4)	0.735 (38.7)	0.75 (38.9)
HPMCP	0.03 (1.8)	0.03 (1.6)	0.03 (1.6)	0.03 (1.6)

[0304] The excess NaOH concentrations for these four ibuprofen gels, were calculated as described in Example 2, and the pH of the gels was directly measured using a pH meter. The pH of the ibuprofen gel (determined using the procedures of the previous examples) increased from 6.58 to 12.22 when the calculated excess NaOH concentration in the gel was increased from 0% to 1.74%. The skin irritation could be related to the pH of the gel, which depends on the excess NaOH concentration.

TABLE 14

Excess NaOH Concentration (wt %) and pH				
	Ibu-1	Ibu-2	Ibu-3	Ibu-4
Excess NaOH Concentration	—	0%	0.98%	1.74%
pH	4.57	6.58	11.83	12.22

[0305] The in vitro permeation of ibuprofen through human cadaver skin from these gels was measured in a slightly different manner than as described in the Methods section. Three diffusion cells were used for each formulation. Normal saline was used as the receiver solution. The entire receiver solution was collected and replaced with fresh saline at each time point. The receiver solution collected was analyzed by an HPLC for the concentration of ibuprofen. The cumulative amount of ibuprofen across human cadaver skin was calculated using the measured ibuprofen concentrations in the receiver solutions, which were plotted versus time and are described below.

[0306] The cumulative amount of ibuprofen across human cadaver skin at 24 hours increased from 0.33 mg/cm² to 5.74 mg/cm² when the calculated excess NaOH concentration in the gel was increased from 0% to 1.74% (Ibu-4), i.e., up to about 17-fold more flux was obtained than with the formulation having 0% excess NaOH concentration. The cumulative amount of ibuprofen that permeated across the human cadaver skin at 24 hours from the formulation with an excess NaOH concentration of 0.98% (Ibu-3, 1.12 mg/cm²) is about 3 times higher than that from the formulation with an excess NaOH concentration of 0% (Ibu-2, 0.33 mg/cm²).

Example 6

[0307] An in vitro skin permeation study was conducted using four phenylpropanolamine hydrochloride transdermal systems, designated PPA-5, PPA-6, PPA-7, and PPA-8, the compositions of which are set forth in Table 15. The matrix

patches were prepared and evaluated using the same procedures as set forth in Example 3. The theoretical percent weight for each ingredient after drying (calculated assuming all the volatile ingredients were completely removed during drying) is listed in Table 16.

TABLE 15

Component Weight and Weight Percent Based on Total Solution Weight				
	PPA-5 g (wt %)	PPA-6 g (wt %)	PPA-7 g (wt %)	PPA-8 g (wt %)
PPA-HCl	0.5 (6.7)	0.5 (5.7)	0.5 (5.6)	0.5 (5.5)
Na ₂ CO ₃	0	0.29 (3.3)	0.44 (5.0)	0.74 (8.1)
DI water	1.0 (13.5)	2.0 (23.0)	2.0 (22.6)	2.0 (21.9)
Methyl alcohol	0.5 (6.7)	0.5 (5.7)	0.5 (5.6)	0.5 (5.5)
PG	0.2 (2.7)	0.2 (2.3)	0.2 (2.3)	0.2 (2.2)
HPMC	0.01 (0.1)	0.01 (0.1)	0.01 (0.1)	0.01 (0.1)
Heptane	1.2 (16.2)	1.2 (13.8)	1.2 (13.6)	1.2 (13.1)
PIB adhesive (30% solid)	4 (54.0)	4 (46.0)	4 (45.2)	4 (45.2)

TABLE 16

Weight and Theoretical Weight Percent Based on Dried Film Weight				
	PPA-5 g (wt %)	PPA-6 g (wt %)	PPA-7 g (wt %)	PPA-8 g (wt %)
PPA-HCl	0.5 (26.2)	0.5 (22.7)	0.5 (21.3)	0.5 (18.9)
Na ₂ CO ₃	0	0.29 (13.2)	0.44 (18.7)	0.74 (27.9)
PG	0.2 (10.5)	0.2 (9.1)	0.2 (8.5)	0.2 (7.5)
HPMC	0.01 (0.5)	0.01 (0.5)	0.01 (0.4)	0.01 (0.4)
PIB adhesive (30% solid)	1.2 (62.8)	1.2 (54.5)	1.2 (51.1)	1.2 (45.3)

[0308] Since PPA-HCl is a salt of a free base, it reacts with Na₂CO₃. The concentration of Na₂CO₃ in the system after the reaction is completed depends on the amount of PPA-HCl added. The remaining sodium carbonate concentration after the reaction is completed is defined as “excess Na₂CO₃ concentration,” which is defined by the following equation.

$$[\text{Na}_2\text{CO}_3 \text{ excess}] = [\text{Na}_2\text{CO}_3 \text{ total}] - [\text{Na}_2\text{CO}_3 \text{ needed for neutralization}]$$

The excess Na₂CO₃ for these four PPA-HCl systems was calculated, and the pH was measured as described in the Methods section. The pH of the PPA-HCl patch increased from 9.81 to 10.17 when the calculated excess Na₂CO₃ concentration in the dried patch was increased from 0.4% to 16.7%.

TABLE 17

Excess Na ₂ CO ₃ Concentration (wt %) and pH				
	PPA-5	PPA-6	PPA-7	PPA-8
Excess Na ₂ CO ₃ Concentration	—	0.4%	6.7%	16.7%
pH	6.54	9.81	9.86	10.17

[0309] The cumulative amount of PPA-HCl across human cadaver skin was calculated using the measured PPA-HCl concentrations in the receiver solutions.